

USB

Design Specification

Ver 3.0

DM270

Digital Still Camera

Department: WW DSC PROJECT

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HISTORY

Version	Date	Author	Notes
Ver 0.1	09/11/99	Masaki Kato	Initial Release for DSC21.
Ver 0.2	09/24/99	Masaki Kato	Updated EP0ICTL and EP0IWDT register description.
Ver 0.3	09/30/99	Masaki Kato	Updated EP0IWDT register description. Updated the description of the setting order of FIFO enable bit and the null/short packet of OUT transaction in 5.4 FIRMWARE PROGRAMMING GUIDE.
Ver 0.4	10/15/99	Masaki Kato	Some comments on Resume bit. Updated the description of the stall and the enable bit of EP0OCTL and EP0ICTL registers. 1. Automatically cleared the enable bit after a transaction. 2. Don't set the stall bit and the enable bit simultaneously.
Ver 0.5	12/06/99	Masaki Kato	Updated for the ES2 circuit. The changes of ES2 are as follows; ☐ Selectable FIFO ENABLE bit automatic clearing scheme ☐ No need to set EP0 OUT ENABLE bit for the setup stage in the control transfer. ☐ Add USB Bus Reset interrupt. ☐ Fixed the width of EP*IRQS registers ☐ Compliant USB Revision 1.1 According to the above changes, updated the description in 5.4 FIRMWARE PROGRAMMING GUIDE.
Ver 0.6	12/15/99	Masaki Kato	Fixed the description error of EP0 IN and EP1 IN ENABLE bit auto clearing scheme. Changed the maximum size of the EP1 FIFO.
Ver 0.7 (Ver 1.0)	08/08/00	Masaki Kato	Compliant with DSC21F(ES3.1). The changes in F-ver are ☐ Asserting SETUP interrupt on ClearFeature(HALT). ☐ Automatic clearing STALL bit when received ClearFeature(HALT). According to the above changes, updated the description of STALL bit of EP1O, EP1I and EP3I, and.
Ver 1.1	09/27/00	Masaki Kato	Release for DSC24(HOKUSAI)/DSC23(DOLPHIN). ☐ Added ClearFeature interrupt. ☐ Added RemoteWakeupStatus. ☐ Added rate feedback mode feature for EP2.
Ver 2.0	05/15/01	Masaki Kato	Release for DSC25(PICASSO). ☐ 16bit access to the EP*ORDT and EP*IWDT registers. ☐ Added DMA feature. ☐ Removed ATTACH related function (status and interrupt, and auto-setting USBRST bit at ATTACH events) ☐ Moved ENCLRMODE bit from EP*ST register to EP*CTL register. ☐ Moved RFB(RateFeedBack Mode) from EP2IST to USBCTL register newly defined.
Ver 2.1	05/21/01	Masaki Kato	☐ Allow ARM register access to the endpoints other than DMA target during DMA.
Ver 2.2	06/13/01	Masaki Kato	☐ Added comments on DMA transfer size. ☐ Updated the block diagram.
Ver 2.3	06/28/01	Masaki Kato	☐ Added DMATRG register and moved DMA start bit (DON) to this register. DMA will start only if DMA transfer size(DSZ) is greater than zero and FIFO available byte size (EP*REST) is also greater than zero.
Ver 2.4	10/29/01	Masaki Kato	☐ Added 4.5 DMA controller section. ☐ Removed description of SETUP interrupt assertion on ClearFeatre(HALT). This function is no longer support because the interrupt status bit is already made for ClearFeature(HALT).
Ver 3.0	03/08/02	Masaki Kato	Release for DM270. ☐ The register area is expanded from 64 half-words to 128. ☐ Added DMA automatic multi-packet transmit function. ☐ Added a new endpoint, ISO IN EP4.

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1. ABSTRACT

This document is described about the USB module for **DM270**. The features of the USB module are as follows;

- Compliant with USB Specification Revision 1.1.
- Involve UDC (USB Device Controller), the Protocol Engine by inSilicon.
- Support only for Full Speed Device.
- Built-in USB Xceiver.
- Built-in 2K RAM for Endpoint FIFO.
- Programmable MaxPktSize.
- Understanding and Decoding of Standard Rquests Except For GetDescriptor, SetDescriptor and SynchFrame.
- Support of Device Remote Wakeup Feature.
- DMA feature.

2. REFERENCES

- Univesal Serial Bus Specification Revision 1.1
- Mass Storage Class Control/Bulk/Interrupt (CBI) Transport Rev 1.0
- Still Image Capture Device Definition Revision 1.0

Above documents can be downloaded from <http://www.usb.org>

- USB Device Controller (UDC) User's Manual Revision 2.0, Phoenix Technology Ltd.
- This document is confidential information under NDA between TI and Phoenix.

3. GLOSSARY

USB	Universal Serial Bus
UDC	USB Device Controller
MaxPktSize	Max Packet Size

4. FUNCTION

4.1 Block Diagram

Figure 1 shows the block diagram of USB module. USB module has a built-in xceiver and 2K bytes RAM. USB module has 7FIFOs and each FIFO is associated each Endpoint. The structure of each FIFO is same except for the direction and the buffer size. The maximum buffer size is 256 bytes for EP0 and EP2 while it is 2048 bytes for EP1, EP3 and EP4. Each FIFO's buffer is allocated onto a unified RAM. The base address and buffer size can be programmed by user.

ARM accesses FIFO read/write port registers to read from or write to FIFOs. DMA interface can be used to transfer data to or from a single FIFO. During a DMA transfer, ARM register access to the same endpoint as the DMA target is not allowed.

ARM clock frequency must be faster than 6MHz. No upper limit.

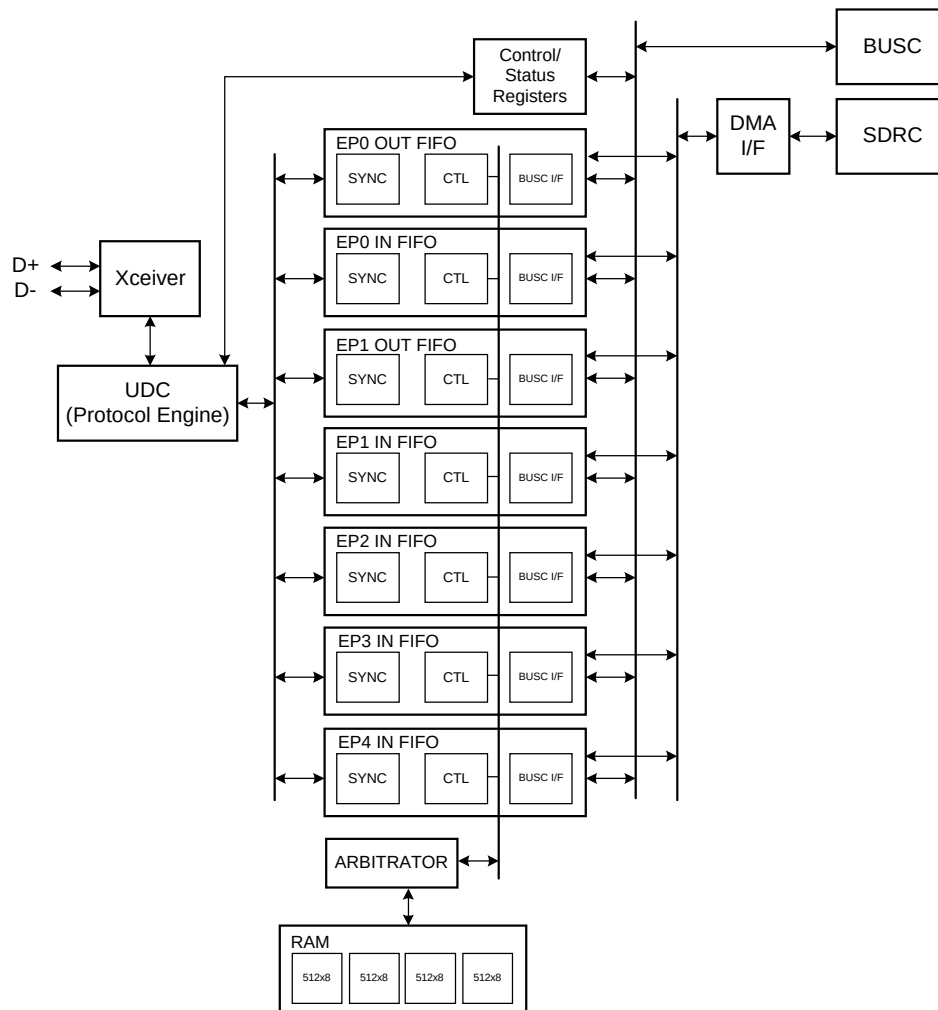


Figure 1 : Block Diagram of USB

4.2 Device Configuration

Figure 2 shows the configuration which USB module supports. There are two configurations which are supported in this module. CfgA supports CBI (Control-Bulk-Interrupt) Transport defined in Mass Storage Class. CfgB is optionally provided to support the configuration defined in “Still Image Capture Device Definition”.

User can configure the following property at device initialization;

1. Whether or not to use each Configuration
2. Each configuration number
3. Whether or not to use each Endpoint for each AlternateSetting
4. MaxPktSize of each Endpoint for each AlternateSetting

The above device initialization is done by writing the initialization data to USBCFG register. It must be done whenever the cable is attached. The format of the actual initialization data will be described in the next subsection.

User cannot change the number of Interface, Interface value, the maximum number of AlternateSettings of each Interface, Endpoint number and Endpoint type.

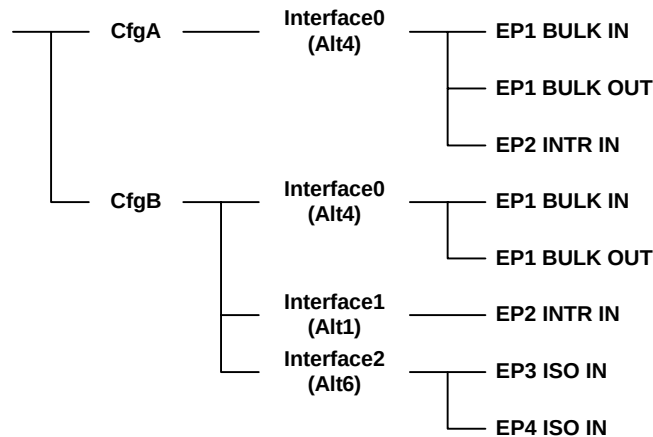


Figure 2 : Device Configuration

4.3 Device Initialization

This section describes the initialization data format. The length of initialization data is 26 half-words. User must write 26 half-words of the data from #0 to #25 to USBCFG register.

Setting of Configuration and Endpoint0

[# 0]

Bits	Description
15-14	Configuration Number of CfgA
13-12	Configuration Number of CfgB
11-10	reserved
9-0	MaxPktSize of EP0

Configuration number can be “00”, “01” or “10”. The configuration set to “00” is not used in the device.

Setting of Endpoints in CfgA

- [# 1 Intf0-Alt0-EP1]
- [# 2 Intf0-Alt1-EP1]
- [# 3 Intf0-Alt2-EP1]
- [# 4 Intf0-Alt3-EP1]

Bits	Description
15	Enable for EP1 IN (1:enable, 0:disable)
14	Enable for EP1 OUT (1:enable, 0:disable)
13-10	reserved

Bits	Description
9-0	MaxPktSize of EP1

If the enable bit for an endpoint is set to 0, the functionality of the specified endpoint is disabled. Device does not respond to the transaction for the disabled endpoint.

- [# 5 Intf0-Alt0-EP2]
- [# 6 Intf0-Alt1-EP2]
- [# 7 Intf0-Alt2-EP2]
- [# 8 Intf0-Alt3-EP2]

Bits	Description
15	Enable for EP2 IN (1:enable, 0:disable)
14-10	reserved
9-0	MaxPktSize of EP2

Setting of Endpoints in CfgB

- [# 9 Intf0-Alt0-EP1]
- [# 10 Intf0-Alt1-EP1]
- [# 11 Intf0-Alt2-EP1]
- [# 12 Intf0-Alt3-EP1]

Bits	Description
15	Enable for EP1 IN (1:enable, 0:disable)
14	Enable for EP1 OUT (1:enable, 0:disable)
13-10	Reserved
9-0	MaxPktSize of EP1

- [# 13 Intf1-Alt0-EP2]

Bits	Description
15	Enable for EP2 IN (1:enable, 0:disable)
14-10	reserved
9-0	MaxPktSize of EP2

- [# 14 Intf2-Alt0-EP3]
- [# 15 Intf2-Alt1-EP3]
- [# 16 Intf2-Alt2-EP3]
- [# 17 Intf2-Alt3-EP3]
- [# 18 Intf2-Alt4-EP3]
- [# 19 Intf2-Alt5-EP3]

Bits	Description
15	Enable for EP3 IN (1:enable, 0:disable)
14-10	reserved
9-0	MaxPktSize of EP3

- [# 20 Intf2-Alt0-EP4]
- [# 21 Intf2-Alt1-EP4]
- [# 22 Intf2-Alt2-EP4]
- [# 23 Intf2-Alt3-EP4]
- [# 24 Intf2-Alt4-EP4]
- [# 25 Intf2-Alt5-EP4]

Bits	Description
15	Enable for EP4 IN (1:enable, 0:disable)
14-10	reserved
9-0	MaxPktSize of EP4

4.4 Power Management

When USB bus is in suspend, all the clocks used in USB module except a clock used for resume detection can be cut-off by register for power saving. The clocks will be automatically activated as soon as the resume signal is detected.

If USB is not used, user can completely disable all the USB clocks by a register in CLKC module. In that case, more power saving can be achieved by connecting VDD pins for USB I/O to GND.

4.5 DMA Controller

Internal DMA controller can be used to transfer data between SDRAM and USB endpoints without ARM intervention.

A uni-direction DMA transmission from/to one endpoint FIFO is allowed in one time. User has to choose one endpoint as DMA target endpoint preceding DMA transfer. User should also specify SDRAM base address and transfer length. Then writing '1' to DON bit in DMATRG register starts DMA transmission. After the transmission of the specified amount of data is completed, ARM is notified the DMA completion by an interrupt.

For OUT endpoints, data stored in OUT FIFO can be transferred to SDRAM. If no data is received in FIFO, setting DON bit does not start DMA. Also the number of data in OUT FIFO is less than DMA transfer length specified by user, the actual DMA transfer length is automatically clipped to actual length.

For IN endpoints, data to be sent to HOST can be transferred from SDRAM. If there is no space for the IN FIFO, setting DON bit has no effect. Also the room for the FIFO is less than the specified DMA length, the actual DMA transfer is clipped by the actual room length.

User can also change DMA endian. Options are LSByte first, MSByte first or byte swap in half-word.

5. REGISTERS

5.1 ARM Memory Map

USB registers are mapped onto the ARM memory area from **0x0003:0900** to **0x0003:09FF**. Offset address shown in the register table in the following subsections means the offset from the USB base address (**0x0003:0900**) with half-word access.

5.2 Register List

The following table shows the list of the registers. Since the bit width of all registers is 16, ARM should access to USB registers width half-word access.

Table 1 : USB Register List

Register Name	Offset address	Description
EP0 OUT Registers		
EP0OAD	0x00	EP0 OUT RAM Base Address
EP0OSZ	0x01	EP0 OUT Buffer Size
EP0OCTL	0x02	EP0 OUT Control
EP0OIRQSZ	0x03	EP0 OUT Threshold for Interrupt
RSV	0x04	Reserved
EP0ORDT	0x05	EP0 OUT FIFO Read Port
EP0OREST	0x06	EP0 OUT Available Byte Size for Read
EP0OST	0x07	EP0 OUT Status
EP0 IN Registers		
EP0IAD	0x08	EP0 IN RAM Base Address
EP0ISZ	0x09	EP0 IN Buffer Size
EP0ICTL	0x0A	EP0 IN Control
EP0IIRQSZ	0x0B	EP0 IN Threshold for Interrupt
EP0IWDT	0x0C	EP0 IN FIFO Write Port
RSV	0x0D	Reserved
EP0IREST	0x0E	EP0 IN Available Byte Size for Write
EP0IST	0x0F	EP0 IN Status
EP1 OUT Registers		
EP1OAD	0x10	EP1 OUT RAM Base Address
EP1OSZ	0x11	EP1 OUT Buffer Size
EP1OCTL	0x12	EP1 OUT Control
EP1OIRQSZ	0x13	EP1 OUT Threshold for Interrupt
RSV	0x14	Reserved
EP1ORDT	0x15	EP1 OUT FIFO Read Port
EP1OREST	0x16	EP1 OUT Available Byte Size for Read
EP1OST	0x17	EP1 OUT Status
EP1 IN Registers		
EP1IAD	0x18	EP1 IN RAM Base Addr
EP1ISZ	0x19	EP1 IN Buffer Size
EP1ICTL	0x1A	EP1 IN Control
EP1IIRQSZ	0x1B	EP1 IN Threshold for Interrupt
EP1IWDT	0x1C	EP1 IN FIFO Write Port
RSV	0x1D	Reserved
EP1IREST	0x1E	EP1 IN Available Byte Size for Write
EP1IST	0x1F	EP1 IN Status
EP2 IN Registers		
EP2IAD	0x20	EP2 IN RAM Base Address
EP2ISZ	0x21	EP2 IN Buffer Size
EP2ICTL	0x22	EP2 IN Control
EP2IIRQSZ	0x23	EP2 IN Threshold for Interrupt
EP2IWDT	0x24	EP2 IN FIFO Write Port
RSV	0x25	Reserved
EP2IREST	0x26	EP2 IN Available Byte Size for Read

Register Name	Offset address	Description
EP2IST	0x27	EP2 IN Status
EP3 IN Registers		
EP3IAD	0x28	EP3 IN RAM Base Address
EP3ISZ	0x29	EP3 IN Buffer Size
EP3ICTL	0x2A	EP3 IN Control
EP3IIRQSZ	0x2B	EP3 IN Threshold for Interrupt
EP3IWDT	0x2C	EP3 IN FIFO Write Port
RSV	0x2D	Reserved
EP3IREST	0x2E	EP3 IN Available Byte Size for Read
EP3IST	0x2F	EP3 IN Status
EP4 IN Registers		
EP4IAD	0x30	EP4 IN RAM Base Address
EP4ISZ	0x31	EP4 IN Buffer Size
EP4ICTL	0x32	EP4 IN Control
EP4IIRQSZ	0x33	EP4 IN Threshold for Interrupt
EP4IWDT	0x34	EP4 IN FIFO Write Port
RSV	0x35	Reserved
EP4IREST	0x36	EP4 IN Available Byte Size for Read
EP4IST	0x37	EP4 IN Status
Misc Registers		
USBRST	0x40	Soft Reset
USBRSM	0x41	Remote Wakeup/Clock Control
USBINTMSK	0x42	Interrupt Enable
USBCFG	0x43	Device Initialization Data Write Port
USBST	0x44	Status
USBINTST	0x45	Interrupt Status
USBFRM	0x46	Frame Number
USBALT	0x47	Configuration/AltSet
USBCTL	0x48	Other Control
DMA Registers		
DMAAD0	0x50	DMA Start Address Low
DMAAD1	0x51	DMA Start Address High
DMASZ	0x52	DMA Transfer Packet Length
DMACTL	0x53	DMA Control
DMATRG	0x54	DMA Start
DMAREST	0x55	DMA Transfer Rest Packet Number
DMAST	0x56	DMA Status

5.3 Detailed Register

5.3.1 EP0 OUT Registers

EP0OAD (0003:0900) Offset : 0x0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	O0AD10	O0AD9	O0AD8	O0AD7	O0AD6	O0AD5	O0AD4	O0AD3	O0AD2	O0AD1	O0AD0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserve.
10:0	O0AD10-O0AD0	EP0 OUT RAM Base Address. Specify the base address of RAM allocated for EP0 OUT FIFO. Setting bit0/1 has no effect.

EP0OSZ (0003:0902) Offset : 0x1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	O0SZ7	O0SZ6	O0SZ5	O0SZ4	O0SZ3	O0SZ2	O0SZ1	O0SZ0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	RSV	Reserve.
7:0	O0SZ7-O0SZ0	EP0 OUT Buffer Size. Specify the buffer size of EP0 OUT FIFO. The actual buffer size equals to this register value added 1. The maximum size of EP0 OUT is 256.

EP0OCTL (0003:0904) Offset : 0x2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	O0EMDE	O0BYTE	O0ENDI	RSV	RSV	RSV	RSV	RSV	O0STL	O0RST	O0ENA
R/W	-	-	-	-	-	R/W	R/W	R/W	-	-	-	-	-	R/W	W	R/W
Default	-	-	-	-	-	0	0	0	-	-	-	-	-	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserve.
10	O0EMDE	FIFO ENABLE bit automatic clear mode select bit. (See the description of bit0 of EP0OCTL register)
9	O0BYTE	Byte access. When '1', data can be read with one byte.
8	O0ENDI	Endian. '0' if little endian, '1' if big endian when reading EP0ORDT.
7:3	RSV	Reserve.
2	O0STL	STALL bit for EP0 OUT. USB specifies the protocol stall in addition to the function stall of Endpoint by SetFeature command. The protocol stall is that when the data packet of the received setup transaction is illegal or unsupported command, the IN/OUT transaction of the successive data or status stage is stalled by the device. When set to one, EP0 OUT is stalled. The device returns a STALL handshake to the HOST after an OUT token is received. Since this bit is automatically cleared when the device receives the next SETUP token, users do not have to clear this bit. Note that do not set O0STL and O0ENA simultaneously.
1	O0RST	RESET bit for EP0 OUT FIFO. When set to one, the status and the data stored in the EP0 OUT FIFO are cleared. This bit is automatically cleared and write only. EP0OAD, EP0OSZ and EP0OIRQSZ are not cleared by this register. When a SETUP token is received, this bit is set automatically and the FIFO is reset to receive the DATA packet without a failure.

Bits	Register Name	Description
0	O0ENA	ENABLE bit for EP0 OUT. When this bit is set, the device returns an ACK handshake after receiving a DATA packet from HOST in OUT transactions to EP0. When this bit is cleared, the device receives but ignores the DATA packet, and returns a NAK handshake. This bit is cleared by hardware when - the DEVICE sends an ACK handshake to the HOST (O0EMDE = 0) - the number of data sent from the HOST in the transaction is less than MaxPktSize when the DEVICE sends an ACK handshake to the HOST (O0EMDE = 1) The enable bit auto clearing scheme can be selected by setting O0EMDE register (bit10 of EP0OCTL register). As for the SETUP transaction, after receiving a SETUP token the device always receives a DATA0 packet and returns an ACK handshake regardless of this bit.

EP0OIRQSZ (0003:0906) Offset : 0x3

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	O0IRQ8	O0IRQ7	O0IRQ6	O0IRQ5	O0IRQ4	O0IRQ3	O0IRQ2	O0IRQ1	O0IRQ0
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:9	RSV	Reserve.
8:0	O0IRQ8-O0IRQ0	EP0 OUT Threshold for Interrupt. Specify the threshold value for the interrupt assertion. (See the description of USBINTST register)

EP0ORDT (0003:090A) Offset : 0x5

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	O0ORDT15	O0ORDT14	O0ORDT13	O0ORDT12	O0ORDT11	O0ORDT10	O0ORDT9	O0ORDT8	O0ORDT7	O0ORDT6	O0ORDT5	O0ORDT4	O0ORDT3	O0ORDT2	O0ORDT1	O0ORDT0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	O0ORDT15-O0ORDT0	EP0 OUT Read Port. Read port of EP0 OUT FIFO. The data field of the received data packet is stored in the FIFO. Read only.

EP0OREST (0003:090C) Offset : 0x6

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	O0OREST8	O0OREST7	O0OREST6	O0OREST5	O0OREST4	O0OREST3	O0OREST2	O0OREST1	O0OREST0
R/W	-	-	-	-	-	-	-	R	R	R	R	R	R	R	R	R
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:9	RSV	Reserve.
8:0	O0OREST8-O0OREST0	EP0 OUT Available Byte Size. Indicate the available byte size for read from EP0 OUT FIFO.

EP0OOST (0003:090E) Offset : 0x7

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	O0FUL	O0EMP
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R	R
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	1

Bits	Register Name	Description
14:2	RSV	Reserve.
1	O0FUL	EP0 OUT Full. Set when the FIFO is full. Cleared when the FIFO is not full.
0	O0EMP	EP0 OUT Empty. Set when the FIFO is empty. Cleared when the FIFO is not empty.

5.3.2 EP0 IN Registers

EP0IAD (0003:0910) Offset : 0x8

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	I0AD10	I0AD9	I0AD8	I0AD7	I0AD6	I0AD5	I0AD4	I0AD3	I0AD2	I0AD1	I0AD0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserve.
10:0	I0AD10-I0AD0	EP0 IN RAM Base Address. Same as EP0OAD.

EP0ISZ (0003:0912) Offset : 0x9

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	I0SZ7	I0SZ6	I0SZ5	I0SZ4	I0SZ3	I0SZ2	I0SZ1	I0SZ0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	RSV	Reserve.
7:0	I0SZ7- I0SZ0	EP0 IN Buffer Size. Same as EP0OSZ. The maximum FIFO size is 256.

EP0ICTL (0003:0914) Offset : 0xA

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	I0EMDE	I0BYTE	I0ENDI	RSV	RSV	RSV	RSV	RSV	I0STL	I0RST	I0ENA
R/W	-	-	-	-	-	R/W	R/W	R/W	-	-	-	-	-	R/W	W	R/W
Default	-	-	-	-	-	0	0	0	-	-	-	-	-	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserve.
10	I0EMDE	FIFO ENABLE bit automatic clear mode select bit. (See the description of bit0 of EP0OCTL register)
9	I0BYTE	Byte access. When '1', data can be written with one byte.
8	I0ENDI	Endian. '0' if little endian, '1' if big endian when writing EP0IWDI.
7:3	RSV	Reserve.
2	I0STL	STALL bit for EP0 IN. When set, the protocol stall condition for EP0 IN is activated. When an IN token to EP0 is received, the device returns a STALL handshake. This bit is automatically cleared after the device receives the next SETUP packet. Note that this stall bit and the enable bit, bit0 of this EP0ICTL, must not be set simultaneously.
1	I0RST	RESET bit for EP0 IN FIFO. When set to one, the status and the data stored in EP0 OUT FIFO are cleared. This register is automatically cleared and write only. EP0IAD, EP0ISZ and EP0IIRQSZ are not cleared by this register.
0	I0ENA	ENABLE bit for EP0 IN. When this bit is set, the DEVICE returns a DATA0/DATA1 packet with data in the FIFO after receiving an IN token to EP0. When this bit is cleared, the DEVICE returns a NAK handshake for IN token. This bit is automatically cleared by hardware when - the DEVICE receives an ACK handshake from the HOST (I0EMDE = 0) - the rest of bytes in FIFO when the DEVICE receives an ACK handshake from the HOST is less than MaxPktSize (I0EMDE = 1) The ENABLE bit control scheme can be selected by setting I0EMDE. If a NAK handshake is received after sending the DATA packet, the data sent to the HOST is recovered. The FIFO status (EP0IST) and this bit are not changed from the status before the transaction.

EP0IIRQSZ (0003:0916) Offset : 0xB

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	I0IRQ8	I0IRQ7	I0IRQ6	I0IRQ5	I0IRQ4	I0IRQ3	I0IRQ2	I0IRQ1	I0IRQ0
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:9	RSV	Reserve.
8:0	I0IRQ8- I0IRQ0	EP0 IN Threshold for Interrupt. Same as EP0IIRQSZ.

EP0IWDT (0003:0918) Offset : 0xC

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	I0WDT15	I0WDT14	I0WDT13	I0WDT12	I0WDT11	I0WDT10	I0WDT9	I0WDT8	I0WDT7	I0WDT6	I0WDT5	I0WDT4	I0WDT3	I0WDT2	I0WDT1	I0WDT0
R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	I0WDT15- I0WDT0	EP0 IN Write Port. Write port of EP0 IN FIFO. The data field of the data packet to be sent to HOST should be stored in the FIFO by writing this register. Write Only.

EP0IREST (0003:091C) Offset : 0xE

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	I0REST8	I0REST7	I0REST6	I0REST5	I0REST4	I0REST3	I0REST2	I0REST1	I0REST0
R/W	-	-	-	-	-	-	-	R	R	R	R	R	R	R	R	R
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	1

Bits	Register Name	Description
15:9	RSV	Reserve.
8:0	I0REST8- I0REST0	EP0 IN Available Byte Size. The available byte size for write to EP0 IN FIFO.

EP0IST (0003:091E) Offset : 0xF

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	I0FUL	I0EMP
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R	R
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	1

Bits	Register Name	Description
15:2	RSV	Reserve.
1	I0FUL	EP0 IN Full.
0	I0EMP	EP0 IN Empty.

5.3.3 EP1 OUT Registers

EP1OAD (0003:0920) Offset : 0x10

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	O1AD10	O1AD9	O1AD8	O1AD7	O1AD6	O1AD5	O1AD4	O1AD3	O1AD2	O1AD1	O1AD0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserve.
10:0	O1AD10- O1AD0	EP1 OUT RAM Base Address.

EP1OSZ (0003:0922) Offset : 0x11

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	O1SZ10	O1SZ9	O1SZ8	O1SZ7	O1SZ6	O1SZ5	O1SZ4	O1SZ3	O1SZ2	O1SZ1	O1SZ0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserve.
10:0	O1SZ10- O1SZ0	EP1 OUT Buffer Size. The maximum FIFO size of EP1 OUT is 2048.

EP1OCTL (0003:0924) Offset : 0x12

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	O1EMDE	O1BYTE	O1ENDI	RSV	RSV	RSV	RSV	RSV	O1STL	O1RST	O1ENA
R/W	-	-	-	-	-	R/W	R/W	R/W	-	-	-	-	-	R/W	W	R/W
Default	-	-	-	-	-	0	0	0	-	-	-	-	-	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserve.
10	O1EMDE	FIFO ENABLE bit automatic clear mode select bit. (See the description of bit0 of EP0OCTL register)
9	O1BYTE	Byte access. When '1', data can be read with one byte.
8	O1ENDI	Endian. '0' if little endian, '1' if big endian when reading EP1ORDT.
7:3	RSV	Reserve.
2	O1STL	STALL bit for EP1 OUT. When set, the function stall condition for EP1 OUT can be applied. On receiving an OUT token with setting O1STL, the internal state of EP1 OUT changes to HALT and the endpoint is stalled. The device receives and ignores the DATA packet, and sends a STALL handshake. Once the endpoint is stalled, the stall condition is not cleared until the HOST clears it by a ClearFeature(HALT) command even if this bit is cleared. If this bit is set with setting O1ENA when ClearFeature(HALT) is received, both bits are automatically cleared at that time. Setting STALL bit without setting O1ENA does not clear them automatically.
1	O1RST	RESET bit for EP1 OUT FIFO. When set to one, the status and the data stored in EP1 OUT FIFO are cleared. This register is automatically clear and write only. EP1OAD, EP1OSZ and EP1OIRQSZ are not cleared by this register.
0	O1ENA	ENABLE bit for EP1 OUT. When this bit is set, the device returns an ACK handshake after receiving the DATA packet from the HOST in the OUT transactions to EP1. When this bit is cleared, the device receives but ignores the DATA packet, and returns a NAK handshake. This bit is cleared by hardware when ☐ the DEVICE sends an ACK handshake to the HOST (O1EMDE = 0) ☐ the number of data sent from the HOST in the transaction is less than MaxPktSize when the DEVICE sends an ACK handshake to the HOST (O1EMDE = 1) The ENABLE bit control scheme can be selected by setting O1EMDE.

EP1OIRQSZ (0003:0926) Offset : 0x13

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	O1IRQ11	O1IRQ10	O1IRQ9	O1IRQ8	O1IRQ7	O1IRQ6	O1IRQ5	O1IRQ4	O1IRQ3	O1IRQ2	O1IRQ1	O1IRQ0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:12	RSV	Reserve.
11:0	O1IRQ11- O1IRQ0	EP1 OUT Threshold for Interrupt.

EP1ORDT (0003:092A) Offset : 0x15

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	O1RDT15	O1RDT14	O1RDT13	O1RDT12	O1RDT11	O1RDT10	O1RDT9	O1RDT8	O1RDT7	O1RDT6	O1RDT5	O1RDT4	O1RDT3	O1RDT2	O1RDT1	O1RDT0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	O1RDT15- O1RDT0	EP1 OUT Read Port.

EP1OREST (0003:092C) Offset : 0x16

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	O1REST11	O1REST10	O1REST9	O1REST8	O1REST7	O1REST6	O1REST5	O1REST4	O1REST3	O1REST2	O1REST1	O1REST0
R/W	-	-	-	-	R	R	R	R	R	R	R	R	R	R	R	R
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:12	RSV	Reserve.
11:0	O1REST11- O1REST0	EP1 OUT Available Byte Size for Read.

EP1OST (0003:092E) Offset : 0x17

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	O1FUL	O1EMP
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R	R
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	1

Bits	Register Name	Description
15:2	RSV	Reserve.
1	O1FUL	EP1 OUT Full. Same as EP0OST.
0	O1EMP	EP1 OUT Empty. Same as EP0OST.

5.3.4 EP1 IN Registers

EP1IAD (0003:0930) Offset : 0x18

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	I1AD10	I1AD9	I1AD8	I1AD7	I1AD6	I1AD5	I1AD4	I1AD3	I1AD2	I1AD1	I1AD0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserve.
10:0	I1AD10- I1AD0	EP1 IN RAM Base Address.

EP1ISZ (0003:0932) Offset : 0x19

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	I1SZ7	I1SZ6	I1SZ5	I1SZ4	I1SZ3	I1SZ2	I1SZ1	I1SZ0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	RSV	Reserve.
7:0	I1SZ7- I1SZ0	EP1 IN Buffer Size. The maximum FIFO size of EP1 IN is 2048.

EP1ICTL (0003:0934) Offset : 0x1A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	I1EMDE	I1BYTE	I1ENDI	RSV	RSV	RSV	RSV	RSV	I1STL	I1RST	I1ENA
R/W	-	-	-	-	-	R/W	R/W	R/W	-	-	-	-	-	R/W	W	R/W
Default	-	-	-	-	-	0	0	0	-	-	-	-	-	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserve.
10	I1EMDE	FIFO ENABLE bit automatic clear mode select bit. (See the description of bit0 of EP0OCTL register)
9	I1BYTE	Byte access. When '1', data can be written with one byte.
8	I1ENDI	Endian. '0' if little endian, '1' if big endian when writing EP1WDT.
7:3	RSV	Reserve.
2	I1STL	EP1 OUT STALL. STALL bit for EP1 IN. When set, the function stall condition for EP1 IN can be applied. On receiving an IN token with setting I1STL, the internal state of the EP1 IN changes to HALT and the EP1 IN is stalled. The device returns a STALL handshake. Once the endpoint is stalled, the stall condition is not cleared until the HOST clears it by a ClearFeature(HALT) command even if this bit is cleared. If this bit is set with setting I1ENA when ClearFeature (HALT) is received, both bits are automatically cleared at that time. Setting I1STL without setting I1ENA does not clear them automatically.
1	I1RST	EP1 OUT FIFO RESET.
0	I1ENA	EP1 OUT ENABLE. ENABLE bit for EP1 IN. When this bit is set, the device returns a DATA0/DATA1 packet with the data in the FIFO after receiving an IN token to EP1. When this bit is cleared, the device returns a NAK handshake for the IN token. This bit is cleared by hardware when - the DEVICE receives an ACK handshake from the HOST (I1EMDE = 0) - the rest of bytes in the FIFO when the DEVICE receives an ACK handshake from the HOST is less than MaxPktSize (I1EMDE = 1) The enable bit control scheme can be selected by setting I1EMDE register (bit10 of EP1ICTL register). If a NAK handshake is received after sending a DATA packet, the data sent to the HOST is recovered, and the FIFO status (EP1IST) and this bit are not changed from the status before the transaction.

EP1IRQSZ (0003:0936) Offset : 0x1B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	I1IRQ11	I1IRQ10	I1IRQ9	I1IRQ8	I1IRQ7	I1IRQ6	I1IRQ5	I1IRQ4	I1IRQ3	I1IRQ2	I1IRQ1	I1IRQ0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:12	RSV	Reserve.
11:0	I1IRQ11-IRQ0	EP1 IN Threshold for Interrupt. Same as EP0IRQSZ except of the bitwidth.

EP1WDT (0003:0938) Offset : 0x1C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	I1WDT15	I1WDT14	I1WDT13	I1WDT12	I1WDT11	I1WDT10	I1WDT9	I1WDT8	I1WDT7	I1WDT6	I1WDT5	I1WDT4	I1WDT3	I1WDT2	I1WDT1	I1WDT0
R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	I1WDT15- I1WDT0	EP1 IN Write Port. Same as EP0IWDT.

EP1IREST (0003:093C) Offset : 0x1E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	I1REST11	I1REST10	I1REST9	I1REST8	I1REST7	I1REST6	I1REST5	I1REST4	I1REST3	I1REST2	I1REST1	I1REST0
R/W	-	-	-	-	R	R	R	R	R	R	R	R	R	R	R	R
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Register Name	Description
15:12	RSV	Reserve.
11:0	I1REST8- I1REST0	EP1 IN Available Byte Size for Write. Same as EP0IREST except of the bitwidth.

EP1IST (0003:093E) Offset : 0x1F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	I1FUL	I1EMP
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R	R
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	1

Bits	Register Name	Description
15:2	RSV	Reserve.
1	I1FUL	EP1 IN Full.
0	I1EMP	EP1 IN Empty.

5.3.5 EP2 IN Registers

EP2IAD (0003:0940) Offset : 0x20

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	I2AD10	I2AD9	I2AD8	I2AD7	I2AD6	I2AD5	I2AD4	I2AD3	I2AD2	I2AD1	I2AD0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserve.
10:0	I2AD10- I2AD0	EP2 IN RAM Base Address. Same as EP0IAD.

EP2ISZ (0003:0942) Offset : 0x21

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	I2SZ7	I2SZ6	I2SZ5	I2SZ4	I2SZ3	I2SZ2	I2SZ1	I2SZ0
R/W	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:8	RSV	Reserve.
7:0	I2SZ7- I2SZ0	EP2 IN Buffer Size. Same as EP0ISZ. The maximum FIFO size of EP2 IN is 256.

EP2ICTL (0003:0944) Offset : 0x22

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	I2EMDE	I2BYTE	I2ENDI	RSV	RSV	RSV	RSV	RSV	I2STL	I2RST	I2ENA
R/W	-	-	-	-	-	R/W	R/W	R/W	-	-	-	-	-	R/W	W	R/W
Default	-	-	-	-	-	0	0	0	-	-	-	-	-	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserve.
10	I2EMDE	FIFO ENABLE bit automatic clear mode select bit. (See the description of bit0 of EP0OCTL register)

Bits	Register Name	Description
9	I2BYTE	Byte access. When '1', data can be written with one byte.
8	I2ENDI	Endian. '0' if little endian, '1' if big endian when writing EP2IWDT.
7:3	RSV	Reserve.
2	I2STL	EP2 IN STALL.
1	I2RST	EP2 IN FIFO RESET.
0	I2ENA	EP2 IN ENABLE.

EP2IIRQSZ (0003:0946) Offset : 0x23

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	I2IRQ8	I2IRQ7	I2IRQ6	I2IRQ5	I2IRQ4	I2IRQ3	I2IRQ2	I2IRQ1	I2IRQ0
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:9	RSV	Reserve.
8:0	I2IRQ8- I2IRQ0	EP2 IN Threshold for Interrupt.

EP2IWDT (0003:0948) Offset : 0x24

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	I2WDT15	I2WDT14	I2WDT13	I2WDT12	I2WDT11	I2WDT10	I2WDT9	I2WDT8	I2WDT7	I2WDT6	I2WDT5	I2WDT4	I2WDT3	I2WDT2	I2WDT1	I2WDT0
R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	I2WDT15- I2WDT0	EP2 IN Write Port.

EP2IREST (0003:094C) Offset : 0x26

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	I2REST8	I2REST7	I2REST6	I2REST5	I2REST4	I2REST3	I2REST2	I2REST1	I2REST0
R/W	-	-	-	-	-	-	-	R	R	R	R	R	R	R	R	R
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	1

Bits	Register Name	Description
15:9	RSV	Reserve.
8:0	I2REST8-REST0	EP2 IN Available Byte Size for Write.

EP2IST (0003:094E) Offset : 0x27

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	I2FUL	I2EMP
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R	R
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	1

Bits	Register Name	Description
15:2	RSV	Reserve.
1	I2FUL	EP2 IN Full.
0	I2EMP	EP2 IN Empty.

5.3.6 EP3 IN Registers

EP3IAD (0003:0950) Offset : 0x28

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	I3AD10	I3AD9	I3AD8	I3AD7	I3AD6	I3AD5	I3AD4	I3AD3	I3AD2	I3AD1	I3AD0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserve.
10:0	I3AD10- I3AD0	EP3 IN RAM Base Address.

EP3ISZ (0003:0952) Offset : 0x29

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	I3SZ10	I3SZ9	I3SZ8	I3SZ7	I3SZ6	I3SZ5	I3SZ4	I3SZ3	I3SZ2	I3SZ1	I3SZ0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserve.
10:0	I3SZ10- I3SZ0	EP3 IN Buffer Size. The maximum FIFO size of EP3 IN is 2048.

EP3ICTL (0003:0954) Offset : 0x2A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	I3BYTE	I3ENDI	RSV	RSV	RSV	RSV	RSV	I3STL	I3RST	I3ENA
R/W	-	-	-	-	-	-	R/W	R/W	R/W	-	-	-	-	-	R/W	W
Default	-	-	-	-	-	-	0	0	0	-	-	-	-	-	0	0

Bits	Register Name	Description
15:11	RSV	Reserve.
10	RSV	Reserve. Writing '1' has no effect.
9	I3BYTE	Byte access. When '1', data can be written with one byte.
8	I3ENDI	Endian. '0' if little endian, '1' if big endian when writing EP3IWDT.
7:3	RSV	Reserve.
2	I3STL	EP3 IN STALL. STALL bit for EP3 IN. On receiving an IN token with setting I3STL, the internal state of the EP3 IN changes to HALT and the endpoint is stalled. The device does not send any data packets to the HOST for IN tokens. Once the endpoint is stalled, the stall condition is not cleared until the HOST clears it by a ClearFeature(HALT) command even if this bit is cleared. If this bit is set with setting I3ENA when ClearFeature(HALT) is received, both bits are automatically cleared at that time. Setting I3STL without setting I3ENA does not clear them automatically.
1	I3RST	EP3 IN FIFO RESET. Same as the bit1 of EP0ICTL.
0	I3ENA	EP3 IN ENABLE. ENABLE bit for EP3 IN. When this bit is set, the device returns a DATA0 packet with the data in the FIFO after receiving an IN token to EP3. When this bit is cleared, the device returns a null DATA0 packet. This bit is not automatically cleared by hardware.

EP3IIRQSZ (0003:0956) Offset : 0x2B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	I3IRQ11	I3IRQ10	I3IRQ9	I3IRQ8	I3IRQ7	I3IRQ6	I3IRQ5	I3IRQ4	I3IRQ3	I3IRQ2	I3IRQ1	I3IRQ0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:12	RSV	Reserve.
11:0	I3IRQ11- I3IRQ0	EP3 IN Threshold for Interrupt.

EP3IWDT (0003:0958) Offset : 0x2C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	I3WDT15	I3WDT14	I3WDT13	I3WDT12	I3WDT11	I3WDT10	I3WDT9	I3WDT8	I3WDT7	I3WDT6	I3WDT5	I3WDT4	I3WDT3	I3WDT2	I3WDT1	I3WDT0
R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	I3WDT15- I3WDT0	EP3 IN Write Port.

EP3IREST (0003:095C) Offset : 0x2E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	I3REST11	I3REST10	I3REST9	I3REST8	I3REST7	I3REST6	I3REST5	I3REST4	I3REST3	I3REST2	I3REST1	I3REST0
R/W	-	-	-	-	R	R	R	R	R	R	R	R	R	R	R	R
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Register Name	Description
15:12	RSV	Reserve.
11:0	I3REST11- I3REST0	EP3 IN Available Byte Size for Write.

EP3IST (0003:095E) Offset : 0x2F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	I3FUL	I3EMP
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R	R
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	1

Bits	Register Name	Description
14:2	RSV	Reserve.
1	I3FUL	EP3 IN Full.
0	I3EMP	EP3 IN Empty.

5.3.7 EP4 IN Registers

EP4IAD (0003:0960) Offset : 0x30

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	I4AD10	I4AD9	I4AD8	I4AD7	I4AD6	I4AD5	I4AD4	I4AD3	I4AD2	I4AD1	I4AD0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserve.
10:0	I4AD10- I4AD0	EP4 IN RAM Base Address.

EP4ISZ (0003:0962) Offset : 0x31

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	I4SZ10	I4SZ9	I4SZ8	I4SZ7	I4SZ6	I4SZ5	I4SZ4	I4SZ3	I4SZ2	I4SZ1	I4SZ0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserve.
10:0	I4SZ10- I4SZ0	EP4 IN Buffer Size. The maximum FIFO size of EP4 IN is 2048.

EP4ICTL (0003:0964) Offset : 0x32

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	I4BYTE	I4ENDI	RSV	RSV	RSV	RSV	RSV	I4STL	I4RST	I4ENA
R/W	-	-	-	-	-	-	R/W	R/W	R/W	-	-	-	-	-	R/W	W
Default	-	-	-	-	-	-	0	0	0	-	-	-	-	-	0	0

Bits	Register Name	Description
15:11	RSV	Reserve.
10	RSV	Reserve. Writing '1' has no effect.
9	I4BYTE	Byte access. When '1', data can be written with one byte.
8	I4ENDI	Endian. '0' if little endian, '1' if big endian when writing EP4IWDT.
7:3	RSV	Reserve.
2	I4STL	EP4 IN STALL. Same as the bit2 of EP3ICTL.
1	I4RST	EP4 IN FIFO RESET. Same as the bit1 of EP0ICTL.

Bits	Register Name	Description
0	I4ENA	EP4 IN ENABLE. Same as the bit0 of EP3ICTL.

EP4IIRQSZ (0003:0966) Offset : 0x33

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	I4IRQ11	I4IRQ10	I4IRQ9	I4IRQ8	I4IRQ7	I4IRQ6	I4IRQ5	I4IRQ4	I4IRQ3	I4IRQ2	I4IRQ1	I4IRQ0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:12	RSV	Reserve.
11:0	I4IRQ11- I4IRQ0	EP4 IN Threshold for Interrupt.

EP4IWDT (0003:0968) Offset : 0x34

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	I4WDT15	I4WDT14	I4WDT13	I4WDT12	I4WDT11	I4WDT10	I4WDT9	I4WDT8	I4WDT7	I4WDT6	I4WDT5	I4WDT4	I4WDT3	I4WDT2	I4WDT1	I4WDT0
R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	I4WDT15- I4WDT0	EP4 IN Write Port.

EP4IREST (0003:096C) Offset : 0x36

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	I4REST11	I4REST10	I4REST9	I4REST8	I4REST7	I4REST6	I4REST5	I4REST4	I4REST3	I4REST2	I4REST1	I4REST0
R/W	-	-	-	-	R	R	R	R	R	R	R	R	R	R	R	R
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Register Name	Description
15:12	RSV	Reserve.
11:0	I4REST11- I4REST0	EP4 IN Available Byte Size for Write.

EP4IST (0003:096E) Offset : 0x37

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	I4FUL	I4EMP
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R	R
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	1

Bits	Register Name	Description
14:2	RSV	Reserve.
1	I4FUL	EP4 IN Full.
0	I4EMP	EP4 IN Empty.

5.3.8 Misc. Registers**USBST (0003:0980) Offset : 0x40**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	MIRST
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
14:1	RSV	Reserve.
0	MIRST	Soft Reset. When set, the protocol engine (UDC) is reset. USBST and USBINTST registers are also reset. This bit is automatically cleared. When the cable is attached, this bit must be set by user.

USBRSM (0003:0982) Offset : 0x41

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	MICKOF	MIRSM
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W	W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0

Bits	Register Name	Description
14:2	RSV	Reserve.
1	MICKOF	Clock Off. Internal clock off control. When set to one, the internal clock is disabled. To enable the clock, clear this bit. This can be used for power saving when USB is in the suspend state. Note that when the device is resumed (including the remote wakeup), this bit is automatically cleared, then the clocks are enabled.
0	MIRSM	Resume. Remote wakeup control register. When USB bus is in the suspend condition, the device can resume the bus by setting this register to one. It is not until DEVICE_REMOTE_WAKEUP feature is enabled by SetFeature command from the HOST that remote wakeup function can be available. At power on reset, this function is disabled and setting this bit has no effects. When set this register, do not set the bit1 of USBRSM to one.

USBINTMSK (0003:0984) Offset : 0x42

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MIDMAIE	MICFIE	MINSPIE	MISOFIE	MISIE	MISCIE	MIBRIE	MISETIE	MISUSIE	MIEP4IIE	MIEP3IIE	MIEP2IIE	MIEP1IIE	MIEP1OIE	MIEP0IIE	MIEP0OIE
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15	MIDMAIE	DMA End Interrupt Enable.
14	MICFIE	ClearFeature Interrupt Enable.
13	MINSPIE	No SOF Packet Interrupt Enable
12	MISOFIE	SOF Interrupt Enable
11	MISIE	SetInterface Interrupt Enable
10	MISCIE	SetConfiguration Interrupt Enable
9	MIBRIE	USB Bus Reset Interrupt Enable
8	MISETIE	Setup Interrupt Enable
7	MISUSIE	Suspend Interrupt Enable
6	MIEP4IIE	EP4 IN Interrupt Enable
5	MIEP3IIE	EP3 IN Interrupt Enable
4	MIEP2IIE	EP2 IN Interrupt Enable
3	MIEP1IIE	EP1 IN Interrupt Enable
2	MIEP1OIE	EP1 OUT Interrupt Enable
1	MIEP0IIE	EP0 IN Interrupt Enable
0	MIEP0OIE	EP0 OUT Interrupt Enable

Interrupt enable register. To enable the interrupt, set each bit to one. To disable the interrupt, clear each bit to zero.

USBCFC (0003:0986) Offset : 0x43

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MICFC15	MICFC14	MICFC13	MICFC12	MICFC11	MICFC10	MICFC9	MICFC8	MICFC7	MICFC6	MICFC5	MICFC4	MICFC3	MICFC2	MICFC1	MICFC0
R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	MICFC15:0	Device Initialization Data Write Port. The initialization is required to make the USB device enable USB transfer. Writing this register must be done in prior to core reset (writing USBRST register). Refer Section 4.3 about the format of initialization data.

USBST (0003:0988) Offset : 0x44

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	MIRMTF	RSV	MISPD	MISETUP	MIDID
R/W	-	-	-	-	-	-	-	-	-	-	-	R	-	R	R	R
Default	-	-	-	-	-	-	-	-	-	-	-	0	-	0	0	0

Bits	Register Name	Description
15:5	RSV	Reserved.
4	MIRMTF	RemoteWakeupFeature Status. Set when the device receives SetFeature(DEVICE_REMOTE_WAKEUP) request. Clear when the device receives ClearFeature(DEVICE_REMOTE_WAKEUP) request.
3	RSV	Reserved.
2	MISPD	Suspend. Suspend status of USB bus. Set when the bus is in the suspend status.
1	MISETUP	Setup. Setup status. Set when the received SETUP transaction requires decoding operation. When the successive IN or OUT token is received, this bit is automatically cleared. Since the device automatically process the standard requests except of GetDescriptor, SetDescriptor and SynchFrame, this bit is not set for these requests. And the data is not stored in EP0 OUT FIFO when the device receives these requests.
0	MIDID	Device Initialize Done. The flag for device initialization done. Set when 20 bytes of the initialization data are written to USBCFG register. Clear at soft reset by USBRST.

USBINTST (0003:098A) Offset : 0x45

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MIDMA	MICFI	MINSPI	MISOFI	MISII	MISCI	MIBRI	MISETI	MISUSI	MIEP4II	MIEP3II	MIEP2II	MIEP1II	MIEP1OI	MIEP0II	MIEP0OI
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15	MIDMA	DMA End Interrupt. Interrupt asserted when DMA transmission is completed.
14	MICFI	ClearFeature Interrupt. Interrupt asserted when ClearFeature(ENDPOINT_HALT) is received.
13	MINSPI	No SOF Packet Interrupt. Interrupt asserted when the device does not receive the next SOF packet within 1ms after a SOF packet is received.
12	MISOFI	SOF Interrupt. Interrupt asserted when a SOF packet is received.
11	MISII	SetInterface Interrupt. Interrupt asserted when SetInterface command is received. AlternateSetting set by HOST can be read from the USBALT register.
10	MISCI	SetConfiguration Interrupt. Interrupt asserted when SetConfiguration command is received. Configuration set by HOST can be read from bit1-0 of USBALT register.
9	MIBRI	USB Bus Reset Interrupt. Interrupt asserted when a USB Bus Reset is received.
8	MISETI	Setup Interrupt. Interrupt asserted when a SETUP transaction whose request is one of GetDescriptor, SetDescriptor and SynchFrame and the class/vendor requests is received.
7	MISUSI	Suspend Change Interrupt. Interrupt asserted when the USB bus is turned into the suspend state or is resumed (including the device remote wakeup). The actual bus status can be known by reading bit2 of USBST register.
6	MIEP4II	EP4I Interrupt. Interrupt asserted when the number of bytes of data stored in FIFO is less than or equals to EP4IIRQSZ at the completion of sending a DATA0 packet.
5	MIEP3II	EP3I Interrupt. Interrupt asserted when the number of bytes of data stored in FIFO is less than or equals to EP3IIRQSZ at the completion of sending a DATA0 packet.
4	MIEP2II	EP2I Interrupt. Interrupt asserted when the number of bytes of data stored in FIFO is less than or equals to EP2IIRQSZ at the completion of receiving of an ACK handshake.
3	MIEP1II	EP1I Interrupt. Interrupt asserted when the number of bytes of data stored in FIFO is less than or equals to EP1IIRQSZ at the completion of receiving of an ACK handshake.
2	MIEP1OI	EP1O Interrupt. Interrupt asserted when the number of bytes of data stored in FIFO is greater than or equals to EP1OIRQSZ or the received DATA packet is short or null at the completion of sending of an ACK handshake.
1	MIEP0II	EP0I Interrupt. Interrupt asserted when the number of bytes of data stored in FIFO is less than or equals to EP0IIRQSZ at the completion of receiving of an ACK handshake.
0	MIEP0OI	EP0O Interrupt. Interrupt asserted when the number of bytes of data stored in FIFO is greater than or equals to EP0OIRQSZ or the received DATA packet is short or null at the completion of sending of an ACK handshake.

Interrupt status registers. The corresponding bit is set when the interruptions occurred. Cleared when this register is read.

USBFRM (0003:098C) Offset : 0x46

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	MIFRN10	MIFRN9	MIFRN8	MIFRN7	MIFRN6	MIFRN5	MIFRN4	MIFRN3	MIFRN2	MIFRN1	MIFRN0
R/W	-	-	-	-	-	R	R	R	R	R	R	R	R	R	R	R
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserve.
10:0	MIFRN10:0	USB Frame number. The latest frame number received from HOST. Reset at USB Bus Reset, a soft reset by USBRST or when the cable is attached or detached.

USBALT (0003:098E) Offset : 0x47

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	MIASI22	MIASI21	MIASI20	RSV	MIASI12	MIASI11	MIASI10	RSV	MIASI02	MIASI01	MIASI00	RSV	RSV	MICONF1	MICONF0
R/W	-	R	R	R	-	R	R	R	-	R	R	R	-	-	R	R
Default	-	0	0	0	-	0	0	0	-	0	0	0	-	-	0	0

Bits	Register Name	Description
15	RSV	Reserved
14:12	MIASI22-MIASI20	AlternateSetting of Interface2. AlternateSetting value set to Interface2 by SetInterface command.
11	RSV	Reserved
10:8	MIASI12-MIASI10	AlternateSetting of Interface1. AlternateSetting value set to Interface1 by SetInterface command.
7	RSV	Reserved
6:4	MIASI02-MIASI00	AlternateSetting of Interface0. AlternateSetting value set to Interface0 by SetInterface command.
3:2	RSV	Reserved
1:0	MICONF1- MICONF0	Configuration. Current Configuration number selected by HOST.

This register indicates the current configuration and alternate setting value of each interface assigned by the HOST. Reset at USB Bus Reset, a soft reset by USBRST. And the alternate setting values (bit14-12, bit10-8, bit6-4) are reset at receiving a SetConfiguration request.

USBCTL (0003:0990) Offset : 0x48

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RFB
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15:1	RSV	Reserved
0	RFB	Rate Feedback Mode. When set to 1, the data toggle bits of EP2 are changed after each data packet without regard to the presence or the type of handshake packet. When 0, EP2 uses the normal data toggle sequence.

5.3.9 DMA Registers

DMAAD0 (0003:09A0) Offset : 0x50

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DAD15	DAD14	DAD13	DAD12	DAD11	DAD10	DAD9	DAD8	DAD7	DAD6	DAD5	DAD4	DAD3	DAD2	DAD1	DAD0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	DAD15-DAD0	DMA start address [15-0]. Only even number is allowed.

DMAAD1 (0003:09A2) Offset : 0x51

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	DAD26	DAD25	DAD24	DAD23	DAD22	DAD21	DAD20	DAD19	DAD18	DAD17	DAD16
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:11	RSV	Reserved
10:0	DAD26-DAD16	DMA start address [26-16].

DMASZ (0003:09A4) Offset : 0x52

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DSZ15	DSZ14	DSZ13	DSZ12	DSZ11	DSZ10	DSZ9	DSZ8	DSZ7	DSZ6	DSZ5	DSZ4	DSZ3	DSZ2	DSZ1	DSZ0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	DSZ15-DSZ0	DMA transfer packet length.

DMACTL (0003:09A6) Offset : 0x53

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	DEP2	DEP1	DEP0	RSV	RSV	DENDI1	DENDI0	RSV	RSV	RSV	RSV	RSV	RSV	RSV	DMDE
R/W	-	R/W	R/W	R/W	-	-	R/W	R/W	-	-	-	-	-	-	-	R/W
Default	-	0	0	0	-	-	0	0	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15	RSV	Reserved
14:12	DEP2-DEP0	DMA Endpoint Select. DEP[2:0] DMA source DMA destination 000 EP0 OUT SDRAM 001 SDRAM EP0 IN 010 EP1 OUT SDRAM 011 SDRAM EP1 IN 100 SDRAM EP2 IN 101 SDRAM EP3 IN 110 SDRAM EP4 IN
11:10	RSV	Reserved
9:8	DENDI1-DENDI0	DMA Endian. DENDI[1:0] 00 D3 D2 D1 D0 -> D3 D2 D1 D0 01 D3 D2 D1 D0 -> D2 D3 D0 D1 10 D3 D2 D1 D0 -> D0 D1 D2 D3 When using "10", DMA start address and transfer length must be the multiple of 4.
7:1	RSV	Reserved
0	DMDE	DMA transmission mode. 0 Automatic multi-packet transmission mode. The transactions with the packet length specified by DSZ register can be achieved automatically. DMA will be terminated by any of the following conditions; - DMAREST becomes 0 - A short packet is received on an OUT endpoint - Clearing DON bit. 1 Single packet transmission mode. DMA will be terminated when a DMA transmission with the length of MaxPacketSize is completed regardless of DSZ register value.

DMATRG (0003:09A8) Offset : 0x54

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	DON
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15:1	RSV	Reserved
0	DON	DMA start. Writing '1' to this bit starts the DMA transfer. Automatically cleared when DMA completes. Clearing this bit terminates the DMA transfer. At the DMA completion, an interrupt is generated. During a DMA transfer, ARM register access to the same endpoint as the DMA target is not allowed. DMA will start only if DMA transfer size(DSZ) is greater than zero and FIFO available byte size (EP*REST) is also greater than zero.

DMAREST (0003:09AA) Offset : 0x55

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DREST15	DREST14	DREST13	DREST12	DREST11	DREST10	DREST9	DREST8	DREST7	DREST6	DREST5	DREST4	DREST3	DREST2	DREST1	DREST0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	DREST15-DREST0	DMA transfer packet rest number. Indicates the rest packet number in the current DMA transmission.

DMAST (0003:09AC) Offset : 0x56

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	DSPKT
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15:1	RSV	Reserved
0	DSPKT	Short packet received status. When the DMA is completed with the condition that an OUT endpoint receives a short/null packet, this bit is set to 1. Cleared when the next DMA is invoked.

6. Interface

Table 2 : USB Top-level I/O List

Pin Name	Bits	I/O	Fr/To	Description
Functional clocking / common signals				
rst_n	1	I	ext	Power on reset..
clk_arm_sys	1	I	clk	ARM clock. Non-gated.
clk_arm	1	I	clk	ARM clock.
clk_usb48	1	I	clk	48MHz xtal clock.
clk_usb12	1	I	clk	12MHz clock input through CTS.
clk_usb12_gated	1	I	clk	12MHz gated clock input through CTS.
clk_usbpll	1	I	clk	PLL clock through CTS.
usb_pllclk	1	O	clk	Internally generated PLL clock to CTS.
usb_clken	1	O	clk	Clock enable.
usb_clk12en	1	O	clk	Clock enable for clk_usb12_gated.
Register access interface signals				
buse_ncs_usb	1	I	buse	Chip select. Active low.
buse_a	6	I	buse	Address input bus.
buse_nrw	1	I	buse	0:read / 1:write.
buse_do	16	I	buse	Data input bus.
usb_do	16	O	buse	Data output bus.
usb_nwait	1	O	buse	Wait.
Interrupt request signals				
usb_nint	1	O	intc	USB interrupt request. Active low.
External interface signals				
usb_rxdp_i	1	I	ext	RXDP of UIS246
usb_rxdm_i	1	I	ext	RXDN of UIS246
usb_rxd_i	1	I	ext	Y of UIS245
usb_txd_o	1	O	ext	TXD of UOS181
usb_se0_o	1	O	ext	SE0 of UOS181
usb_gz_o	1	O	ext	GZ of UOS181
usb_suspend	1	O	tst	Suspend to PWRDN of UIS245.
DMA interface signals				
sd_req	1	O	sdrc	DMA request
sd_ack	1	I	sdrc	DMA acknowledge
sd_addr	27	O	sdrc	DMA address
sd_bcmt	6	O	sdrc	DMA byte count
sd_rnw	1	O	sdrc	DMA read/write
sd_wdata	32	O	sdrc	DMA write data
sd_rdata	32	I	sdrc	DMA read data
Test signals				
tst_scan	1	I	tst	SCAN mode.
tst_pmt	1	I	tst	USB RAM PMT mode.
tst_pmt_a	9	I	tst	PMT A.
tst_pmt_d	8	I	tst	PMT D.
tst_pmt_ez	1	I	tst	PMT EZ.
tst_pmt_wz	1	I	tst	PMT WZ.
tst_pmt_wrenz	8	I	tst	PMT WRENZ.
tst_pmt_q	8	O	tst	PMT Q.
tst_pmt_pass	1	O	tst	PMT pass.

ファイル名 :	dsc27_usb_spec_3_0.doc
フォルダ :	D:\usr\mkato\work\dsc\dsc27\dsn_spec\usb
テンプレート :	C:\Program Files\Microsoft Office\Template\Normal.dot
表題 :	USB Design Specification
副題 :	DM270
作成者 :	Masaki Kato
キーワード :	USB
説明 :	Ver 3.0
作成日時 :	00/09/27 20:45
変更回数 :	80
最終保存日時 :	02/05/28 10:21
最終保存者 :	Texas Instruments Japan Ltd.
編集時間 :	1,024 分
最終印刷日時 :	02/05/28 11:49
最終印刷時のカウント	
ページ数 :	29
単語数 :	8,378 (約)
文字数 :	47,760 (約)